

A Minimal Study of Echo Chambers in Small Networks

Abstract. We study how echo chambers form in small simulated networks of a few dozen nodes.

1 Introduction

Echo chambers reduce exposure to opposing views. Prior work studied large platforms [1]; we ask whether the effect appears in small networks. We simulate opinion dynamics with a bounded-confidence model and measure clustering of opinions over time, following the interaction mechanisms described for social influence [2] and complex networks [3]. Our experiments show that even small networks develop stable opinion clusters when agents preferentially interact with similar peers, which suggests the effect is structural rather than platform-specific. This reading of the mechanism is consistent with the theoretical predictions in [2].

2 Conclusion

We presented a minimal bounded-confidence simulation showing opinion clustering in small networks, consistent with the theoretical predictions in [2]. Future work should test the mechanism with empirical interaction logs.

References

- [1] Cinelli M, et al. The echo chamber effect on social media. PNAS, 2021.
- [2] Deffuant G, et al. Mixing beliefs among interacting agents. Advances in Complex Systems, 2000.
- [3] Watts D J, Strogatz S H. Collective dynamics of small-world networks. Nature, 1998.